

ORIGINAL ARTICLE

Forecasting Consumer Price Index (CPI) - Empirical Analysis from Tanzania

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Citation (APA Style):

Mchukwa, E. W., Mbwambo, S.K., & Mchomvu, Z.B. (2025). Forecasting Consumer Price Index (CPI) - Empirical Analysis from Tanzania. *Economic Insights – Trends and Challenges*, 14(1), 73-83. <https://doi.org/10.51865/EITC.2025.01.05>

Abstract: All nations worldwide including developed and developing countries like Tanzania strive to attain the stabilized cost of goods and services commonly referred as inflation which is measured using Consumer Price Index (CPI). In pursuit of this objective, monetary policies are precisely made to not alter the cost of goods and services in way that will make goods and services expensive for consumers. In this pursuit, this study aims to forecast the costs of goods and services using CPI data as measure of inflation from 1966 to 2021 found from World Bank. In making theoretical argument, Adaptive Expectation Theory (AET) was employed. Further, for empirical analysis, an univariate time series model which is Autoregressive Integrated Moving Average Model (ARIMA) model was applied in which the differenced ARIMA (0,1,0) model was found to be the best model compared to undifferenced ARIMA (0,0,0) model. The findings of this study came to reveal that the inflation rate of Tanzania measured as CPI have gradual declining tendency in the next ten (10) years (decade). These results recommend to policymakers and Central Bank of Tanzania referred as Bank of Tanzania (BOT) to make good monetary policies to make sure that the low inflation rate is attained to reduce the costs of goods and services.

Keywords: Consumer Price Index (CPI); Inflation; ARIMA; Tanzania Commitment.

JEL Classification: E51; E52; E58.

1. Introduction

Inflation signifies a constant increase in the general price level of goods and services within an economy over a given time period (Boniface & Martin, 2019). This often leads to a decline in the buying power of money, with each unit of money now being able to purchase a decreased number of goods and services (Bunjaku et al., 2023) introducing unpredictability into the economy in the borrowing and lending of money to borrowers and lenders (Nyoni, 2019c). Furthermore, individuals with rigid wages, such as pensioners often encounter challenges since inflation causes the reduction of their revenues (Wanjuki et al., 2021).

Globally, the rate of inflation measured by CPI has not been steady, taking 1966 as the base year, in 1987 the rate of inflation was 5.7%, a fall from 12.4% in 1981 which has fallen to 8.3% in 2022 (Aabeyir, 2019). In Sub Saharan Africa, the rate of inflation has not been rigid as well in which the highest ever reached inflation rate as measured by CPI was 27.4% in 1994 and the lowest reached CPI rate was 2.8% in 2019 which fell again to 9.5% in 2022 (Olufunmike et al., 2019) indicating fluctuating inflation rates.

Similarly, Tanzania evidenced the fluctuations in inflation rate as measured using CPI. The highest attained inflation rate in Tanzania was 36.1% in 1984 while the lowest ever inflation rate was 3.3% in 2020 (Olufunmike et al., 2019) and rose again up to 5% in 2023 (Bank of Tanzania (BOT), 2024). This being a problem, the government of United Republic have been doing several money policy actions such as applying contractionary monetary policy in the economy and lowering wages (Olufunmike et al., 2019). However, despite these remarkable efforts, combating inflation still poses a challenge mainly due to the rising demand of the imported goods and services which inflates the domestic economy of Tanzania.

Several studies have examined the fluctuations in the inflation rate in sub-Saharan Africa in different countries such as studies of Aabeyir (2019), Bai et al. (2021), Jere et al. (2019), Mwanga (2020), Olufunmike et al. (2019) and Pillay (2020) and little attention has been given to Tanzania as a separate country.

Therefore, this study aims to forecast the inflation rate of Tanzania measured by Consumer Price Index (CPI).

Together with this introduction, this study covered literature review which highlighted the studies similar to this study topic, theoretical model, the data source and types of data employed, empirical model, analysis of results, discussion of the findings as well as conclusion.

2. Literature Review

Wati et al. (2021) used Autoregressive Integrated Moving Average (ARIMA) model to forecast the inflation rate in Indonesia. The data on the expenditures of beverages, foods, tobacco and cigarettes from 2010 to 2019 were used in this study. The findings of this study where ARIMA (12,0,12) was used as the best model found the inflation rate which was measured using CPI tends to increase over time. In similar fashion, Marpaung et al. (2022) forecasted the Indonesia's inflation rate using the Box Jenkins method which is commonly referred as ARIMA. Using time series data in monthly basis from January 2016 to April 2021 collected from Central Java Province Statistical Agency, this study revealed that ARIMA (3,0,3) as the best model. Additionally, this study found the increasing trend of inflation.

In Uganda, Mwanga (2020) forecasted CPI using data found from Ugandan Bureau of Statistics for the time period from January 2010 to July 2020 where an ARIMA (1, 1, 1) and (0, 1, 1) with no constant term were chosen as the best models. This study found that Ugandan CPIs tend to fluctuate between 4.7% to 5.0%.

On the contrary, Deka & Resatoglu (2019) examined the best model suitable for forecasting CPI in Turkey using ARIMA model. Data from February 2004 to August 2018 collected from Organization for Economic Cooperation and Development (OECD) was used for study's analysis. The findings of this study came to find that ARIMA (1,1,4) is the best ARIMA model to use in forecasting inflation (CPI). Different from Deka & Resatoglu (2019), Aabeyir (2019) used an exponential smoothing method to forecast the values of CPI in Ghana. In doing so, Minitab software was used to examine the CPI figures over the time period March 2013 to November 2018. This study revealed that Winters' additive method is the best model for forecasting the values of CPIs. Moreover, Bunjaku et al. (2023) employed an ARIMA model in forecasting CPI of Kosovo in Monthly wise to reveal that an ARIMA (3, 1, 2) forecasts found the significant values of CPI which keep on rising. In parallel way, Nyoni (2019a) who used ARIMA (1, 1, 1) as the best model found the upward trajectory inflation for the next decade in German in which data from 1960 to 2017 found from World Bank was examined using the Box – Jenkins (ARIMA) method. Correspondingly, Nyoni (2019b) used ARIMA (1, 2, 1) as the best model found the upward trajectory inflation for the next decade in Iran in which data from 1960 to 2017 found from World Bank was examined using the Box – Jenkins (ARIMA) method.

Furthermore, Wanjuki et al. (2022) used Seasonal Autoregressive Integrated Moving Average (SARIMA) to forecast food and beverages prices using monthly secondary data from January 1991 to February 2020 obtained from Kenyan national bureau of statistics. This study's forecast employed R software where the SARIMA models were compared using Mean Absolute Scaled Error (MASE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). The findings of this study which was obtained from SARIMA (1,1,1) (0,1,1)₁₂ found the positive trajectory trend of the values of inflation for the next 2 years (24 months) which was measured using CPI. Jere et al. (2019) compared the forecasting capacity of ARIMA and Multicointegration (ECM) models in Zambia using monthly data collected from January 2003 to December 2017. This study came to reveal that ECM is the best model for forecasting Zambian CPI patterns.

3. Theoretical Model

Adaptive Expectation Theory (AET)

The adaptive expectations theory was developed by American classical economist John F. Muth in 1960s to outline how individuals and countries make expectations of the forthcoming value of variables based on the previous and present data outlooks such as making future inflation expectations based on the past and current values of inflation (Sutton & Barto, 1981).

The AET posits that individuals and nations which are led by governments change their expectations progressively as time passes by observing the past and present figures. This theory is of crucial importance in forecasting Tanzania's rate of inflation as measured by CPI since it acknowledges the fact that the past and present values often give good insights to predict the future values. Additionally, on knowing the past figures often pave a way to the individuals and governments to make rational decisions based on past experiences. Despite the critique raised by Gertchev (2007) arguing that AET is only favorable to capture short term economic dynamics, this theory is still useful in forecasting the future values of inflation since inflation is not a snap short process as it needs gradual price increment (Singla et al., 2019).

The Data

In forecasting the future values of CPI in Tanzania, secondary world bank data for time period from 1966 to 2021 was used. Despite having gaps in earlier periods, this data source is useful in this study since it is credible for collecting vast amount of data worldwide (Utouh et al., 2024).

Table 1 presents an overview of the descriptive statistical analysis made on CPI data sourced from world bank. The data after descriptive analysis shows that on average CPI value for Tanzania is 15.0463% revealing evidence on the increase in the prices of goods and services for the past 56 years. Moreover, the standard deviation observed of 10.57599% shows how CPI fluctuates from its own mean over the 56 examined years. On range as presented by the minimum and maximum values, CPI tends to range from 3.290291% as the minimum value to the maximum value of 36.14594%.

Table 1. Descriptive statistics of CPI from 1966 to 2021

Variable	Observation	Mean	Std. dev.	Min	Max
CPI	56	15.04637	10.57599	3.290291	36.14594

Source: Authors' computations (2024).

4. Empirical Model

The Autoregressive Integrated Moving Average (ARIMA) model was used in this study for forecasting CPI of Tanzania for the next one decade from 2022 to 2031. The ARIMA model, which is made up of Autoregression (AR) and Moving Average (MA) models is suitable in this study since this study aims to study and predict an univariate time series variable (Boniface & Martin, 2019) which is inflation measured as CPI in this study.

The AR model with order p is expressed as:

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \dots + \phi_p Z_{t-p} + \varepsilon_t, t > p \quad (1)$$

Where:

Z_t = dependent variable at time (t);

$Z_{t-1}, Z_{t-2}, \dots, Z_{t-p}$ are dependent variables at lagged periods;

ϕ_i = coefficients to be estimated at order p ;

ε_t = error term with zero mean and constant variance; $\varepsilon_t \sim N(0, \sigma^2)$.

MA with order q is expressed as:

$$Z_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q} \quad (2)$$

Where:

Z_t = dependent variable at time (t);

ε_t = error term with zero mean and constant variance; $\varepsilon_t \sim N(0, \sigma^2)$;

$\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-q}$ are dependent variables at lagged periods;

θ_i = coefficients of MA to be estimated at order q .

The combination of AR, differenced/integral component and MA with orders p, d, q respectively to produce ARIMA (p, d, q) model is expressed as:

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \dots + \phi_p Z_{t-p} + \dots + d Z_{t-p-d} + \varepsilon_t \quad (3)$$

5. Results/Analysis

This section presents the tests and estimations as well as forecasts made in this study.

5.1. Test for Stationarity

For proper time series analysis, it is crucial to make sure that the dataset is stationary meaning that they should have zero mean and constant variance. Figure 1 presents the findings that the dataset shown properties of non-stationarity, meaning the lack of a zero mean and constant variance in the dataset under study. This is evident from the fact that the data points across all the years under study running from 1966 to 2021 did not show fluctuations around the zero mean as shown in Figure 1.

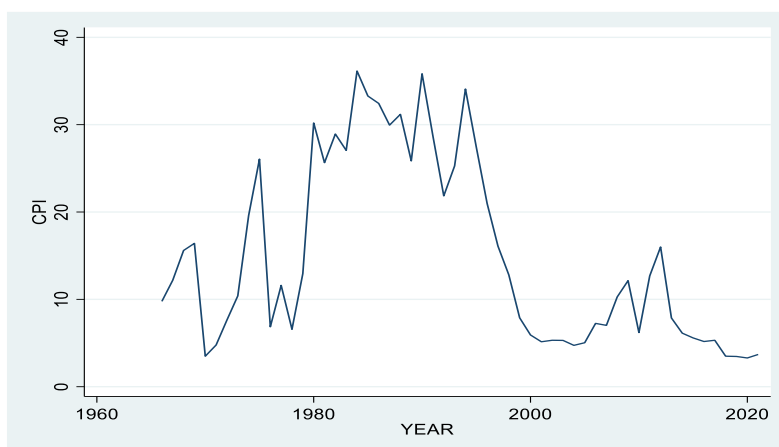


Fig. 1. Non-Stationarity Test Result

Source: Made by authors (2024).

In addressing non-stationarity problem which is likely to cause errors in future analysis of data, differencing method was applied. Upon the application of first difference on CPI, the non-stationarity trend was solved as the data became stationary as presented in both Table 2 and Figure 2. As presented in Table 2, the p-value of 0.2945 suggests that at 5% conventional significance level the null hypothesis of non-stationarity cannot be rejected which in other way it means that there is no sufficient evidence to claim that the series is stationary. However, since the data are fluctuating from the zero-mean it indicates that stationarity assumption holds. Additionally, the Augmented Dickey Fuller test after the first difference had a p-value of 0.000 which is significant at 1% and lower than 5% significance level indicating a strong presence of stationarity which is equal to strong absence of non-stationarity within the data examined.

Table 2. Dickey Fuller Stationarity Test

Variable	P value
d.cpi	0.2945
d.cpi	0.0000***

Where *** indicated significance p value at conventional 5% significance level

Source: Made by authors (2024).

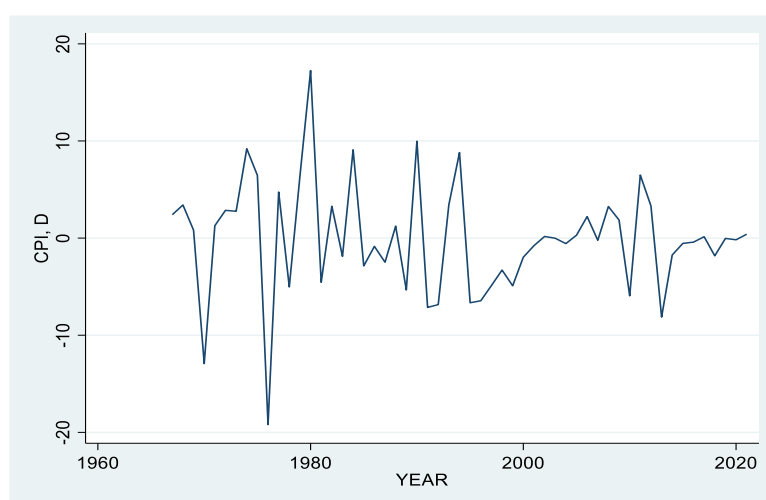


Fig. 2. Stationary Results

Source: Made by authors (2024).

5.2. Autocorrelation Tests

Autocorrelation (serial correlation) assumption also has to be held into account in time series analysis (Olufunmike et al., 2019), meaning that observations have to be different at different time points (Nyoni, 2019c). Figure 3 indicates the results for autocorrelation tests. The data trends indicated that the time series data were serially correlated from time to time which will likely lead to overestimation or underestimation of statistical estimates to be estimated leading to biased, inefficient estimates and wrong conclusions which reduces reliability of the findings.

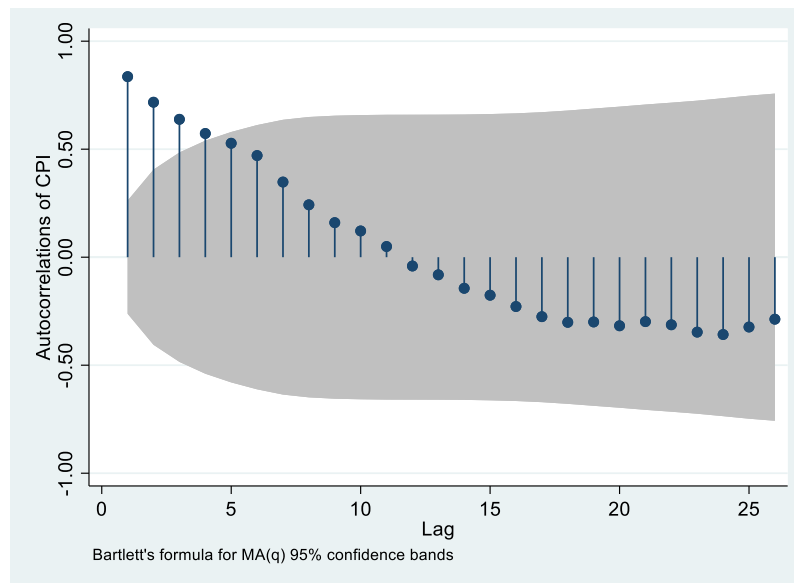


Fig. 3. Autocorrelated Results

Source: Made by authors (2024).

Then, in addressing serial correlation, the differenced CPI was used since it includes lagged variables in estimating the regression model. Figure 4 displays the results for non-autocorrelated data after applying first difference to make it non-autocorrelated.

5.3. Finding the Values of p, d and q

In order to use the correct ARIMA model, knowing Autoregressive (AR) order abbreviated as p, Moving Average (MA) order abbreviated as q and differencing order abbreviated as d is paramount for indicating how many lagged periods should be used for both past values of CPI and error terms (Mwanga, 2020). Therefore, using Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) the p and q orders were identified.

The order of Moving average (q) was similarly found to be zero (0) as shown in Figure 4 whereas the order of differencing (d) was 0 to 1 due to the first difference.

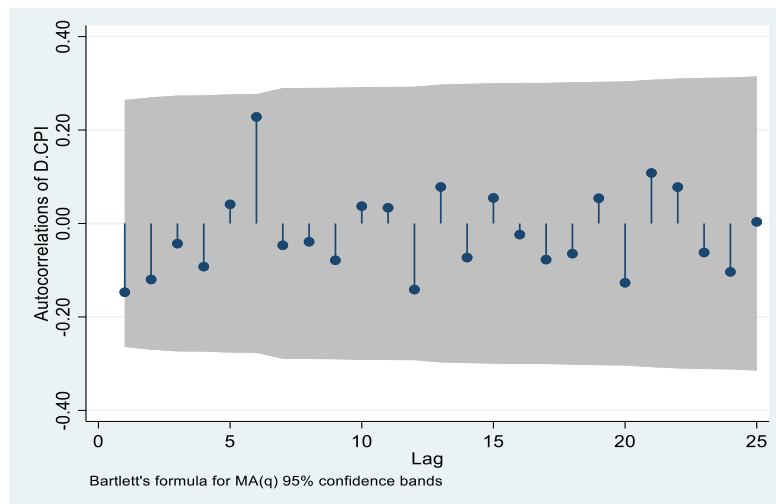


Fig.4. Moving Average Order (q)

Source: Made by authors (2024).

Figure 5 presents the results for order (p) of AR model which suitably fits in this study. As Figure 4 presents, the AR (p) order to use in this study is zero since all values were found to be falling in the shading confidence band.

Therefore, since the study employed the first difference, this study had two competing models ARIMA (0,0,0) and ARIMA (0,1,0).

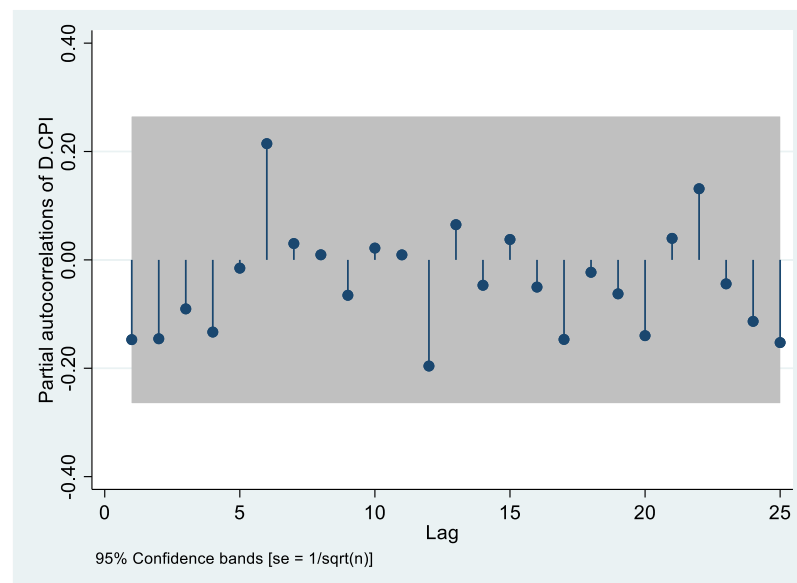


Fig. 5. Autoregressive Order (p)

Source: Made by authors (2024).

5.4. Model Estimations

Akaike and Bayesian Information criterion method was used to estimate and choose the best model for forecasting as shown in Table 3. According to Deka & Resatoglu (2019) the best model is the model with less Sigma SQ indicating less variability or dispersion in the data, higher values of Log Likelihood suggesting the higher probability of making data observations in the statistical parameter hence showing the data fitness and smaller values of Akaike and Bayesian information which indicate better model fitness. Therefore, as Table 3 presents, among the two competing models which are ARIMA (0,0,0) and ARIMA (0,1,0), the ARIMA (0,1,0) is the best model.

Table 3. Best Model Estimation

	ARMA=ARIMA (0,0,0)	ARIMA (0,1,0)	
Sigma SQ (less is good)	10.48114	5.818932	
Log likelihood (Bigger is good)	-211.0369	-174.903	
Akaike (smaller is better)	426.0737	353.8061	
Bayesian (smaller is better)	430.1244	357.8207	
Best model			ARIMA (0,1,0)

Source: Made by authors (2024).

5.5. CPI Forecasting

White noise test for error

Before forecasting using the best-chosen model, checking for white noise for error ensured that errors are less likely to mislead the results and show that the time series is a sequence of random values and cannot be predicted unless with proper time series model which is ARIMA (0,1,0) in this study. Table 4 below indicates that errors are white noise hence the model used for forecasting. The white noise errors are the ones with probability value that is greater than 5% conventional level of significance. Therefore, 0.9346 is greater than 5% significance level indicating that the errors are white noise and therefore cannot mislead the findings and study conclusion.

Table 4. White Noise Results

Criterion	P value
Portmanteau (Q) statistic	15.2816
Prob > chi2(25)	0.9346***

Source: Made by authors (2024).

p*** indicates that p value for error is greater than 5% significance level hence white noise.

Forecasted CPI results

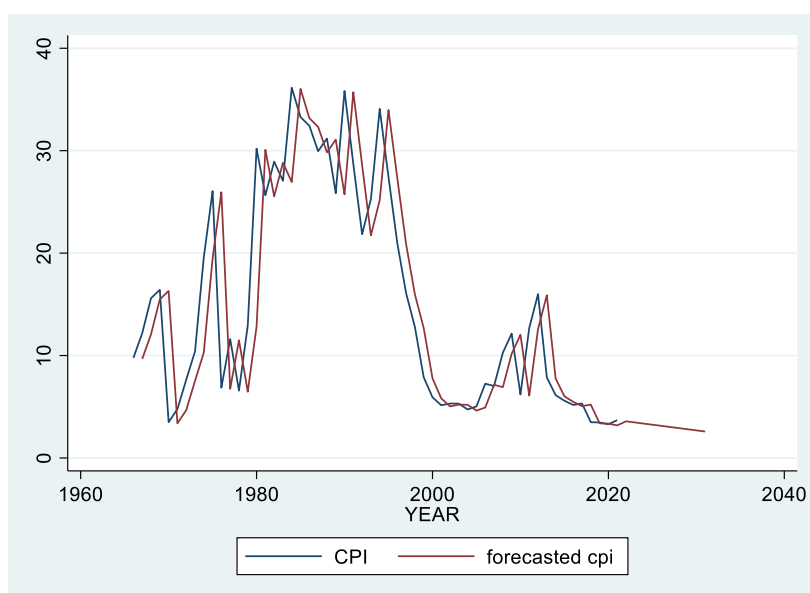
Table 5 indicates the values of forecasted CPIs over ten-year period as forecasted using ARIMA (0,1,0). In 2022, the forecasted CPI is 3.580263% which will decrease to 3.469607% in 2023, which will still keep on decreasing to reach 3.358951% in 2024. Over the ten-time period, the CPI will keep on decreasing to reach 2.58437% which is the lowest level of CPI in the ten (10) forecasted years. This suggests the fact that an increase in prices (positive change) is being reduced more and more, which leads to the conclusion that the tendency is to stabilize prices and reach macroeconomic equilibrium.

Table 5. Tanzania Forecasted CPIs

Year	Forecasted CPI (%)
2022	3.580263
2023	3.469607
2024	3.358951
2025	3.248294
2026	3.137638
2027	3.026982
2028	2.916326
2029	2.805669
2030	2.695013
2031	2.584357

Source: Made by authors (2024).

In a parallel way, as presented in Figure 6, the red line represents the forecasted CPI values; these forecasted CPIs reveal the progressively gradual decrease in CPI of Tanzania across the forecasted 10 years. Thus, over the next decade, the inflation rate of the economy of Tanzania shows the consistent downward declining trend.

**Fig. 6.** Lagged CPI versus Forecasted CPI

Source: Made by authors (2024).

6. Discussion of Findings

This study found that the Consumer Price Index (CPI) for Tanzania which is a common measure of inflation was found to be decreasing gradually over the ten studied years.

The findings of this study are contrary to the studies of Bunjaku et al. (2023) who found that as the days progress, the inflation rate of Kosovo is likely to increase due to year to year supply shocks in the economy, Nyoni (2019a) also found that, inflation rate of Saudi Arabia will keep on rising from time to time, as it is the case with the observation of Nyoni (2019b) who found that the rate of inflation in German will keep getting high from time to time, in similar way Wanjuki et al. (2021) forecasted the value of inflation for food products in Kenya to find the upward trend in the general prices of food and food products in Kenya, in similar fashion Wati

et al. (2021) forecasted CPI of Indonesia to find that the CPI trends keep on getting high from time to time, in cementing this Jere et al. (2019) forecasted the values of CPI in Zambia in which the study came to find that the general price level in Zambia increases from time to time, Boniface & Martin (2019) also forecasted the values of CPI in Ghana to find that increasing trend of CPI, Nyoni (2019c) forecasted the values of CPI in Iran to also find the upward trend of CPI and Marpaung et al. (2022) who did CPI forecasts in Central Java to find the upward CPI trend indicating the general increase in the prices of goods and services in the economy.

In contrary to many studies but in parallel way to this study, Mwanga (2020) found that the CPI tends to have some series of increasing and decreasing depending on the policies made and general global economic conditions.

7. Conclusions

This study used data sourced from the World Bank running from 1966 to 2021, a total of 56 years to forecast the Consumer Price Index (CPI) of Tanzania for the next forthcoming decade (ten years' time period) which runs from 2022 to 2031. To achieve this purpose, this study used an Autoregressive Integrated Moving Average (ARIMA) model as one of the time series models which is purposively used to forecast a univariate time series variable which is CPI in this study.

The findings of the study revealed the reduction of CPI in Tanzania in the next forecasted ten years. Specifically, the rate of inflation, as measured by the CPI, revealed a downward trend, which is approximately decrease to reach 2.58% by the year 2031 from its first forecasted value of 3.58% in 2022. These findings suggest a persistent decline of inflationary pressures in Tanzania implying a reduction in the rise in prices of goods and services over the next forecasted decade.

This study's findings give valuable insights for policymakers to formulate good monetary policies that will contract the amount of money supply in the economy and being mindful about short term and long-term targets in order to achieve economic stability.

Despite the enlightening findings of this study, the usage of CPI data from 1966 to 2021 provides a vast of historical economic outlook but economic conditions which affect CPI might have been changing over the time. Employing this historical data might not include recently economic shocks like Covid 19 eruptions, Russia- Ukraine wars and government policy adjustments such as interest rates, price control mechanisms such as price ceiling and price floor which may impact the general price levels in the consumer basket of goods and services in the economy leading to less accurate forecasts of the CPI of Tanzania.

Additionally, ARIMA models always relies on single time series variable and lacks the incorporation of other exogenous variables which are likely to influence CPI leading to lack precision on estimate forecasts.

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